

Exploratory Data Analysis

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Learning Objectives

By the end of this session, you will be able to:

- Understand the variables in a dataset
- Identify different variable types
- Perform univariate analysis
- Use visualizations to detect patterns
- Identify correlations and outliers

Understand the Variables

Before creating any plot or statistic, ask:

- What variables are available?
- What does each variable represent?
- Is there a target variable?
- Are there suspicious or unclear variables?

Data Dictionary

For each variable, we should know:

- Name
- Data type
- Unit of measurement
- Description
- Possible values

Original vs Derived Variables

Original Variables

- Collected directly from the source

Derived Variables

- Created from other variables
- Example: Age calculated from Date of Birth

Types of Variables

In most datasets, variables fall into a few main categories:

- **Numerical variables**
- **Categorical variables**
- **Binary variables**

Understanding the variable type helps determine:

- the appropriate analysis method
- the correct type of visualization

Numerical Variables

Remember, they can be divided into:

Discrete

- Countable values
- Example: number of purchases

Continuous

- Any value within a range
- Example: height, income, temperature

Categorical Variables

Two common types:

Nominal

- No natural order
- Example: country, product type

Ordinal

- Categories with a meaningful order
- Example: education level, satisfaction rating

Binary Variables

Binary variables contain **only two possible values**.

Examples:

- Yes / No
- True / False
- 0 / 1
- Purchased / Not Purchased

Binary variables are very common in:

- classification problems
- machine learning models

Univariate Analysis

Univariate analysis examines **one variable at a time**.

Main objectives:

- Understand the distribution of the variable
- Identify unusual values
- Summarize key statistics

The analysis depends on the **type of variable**.

Descriptive Statistics

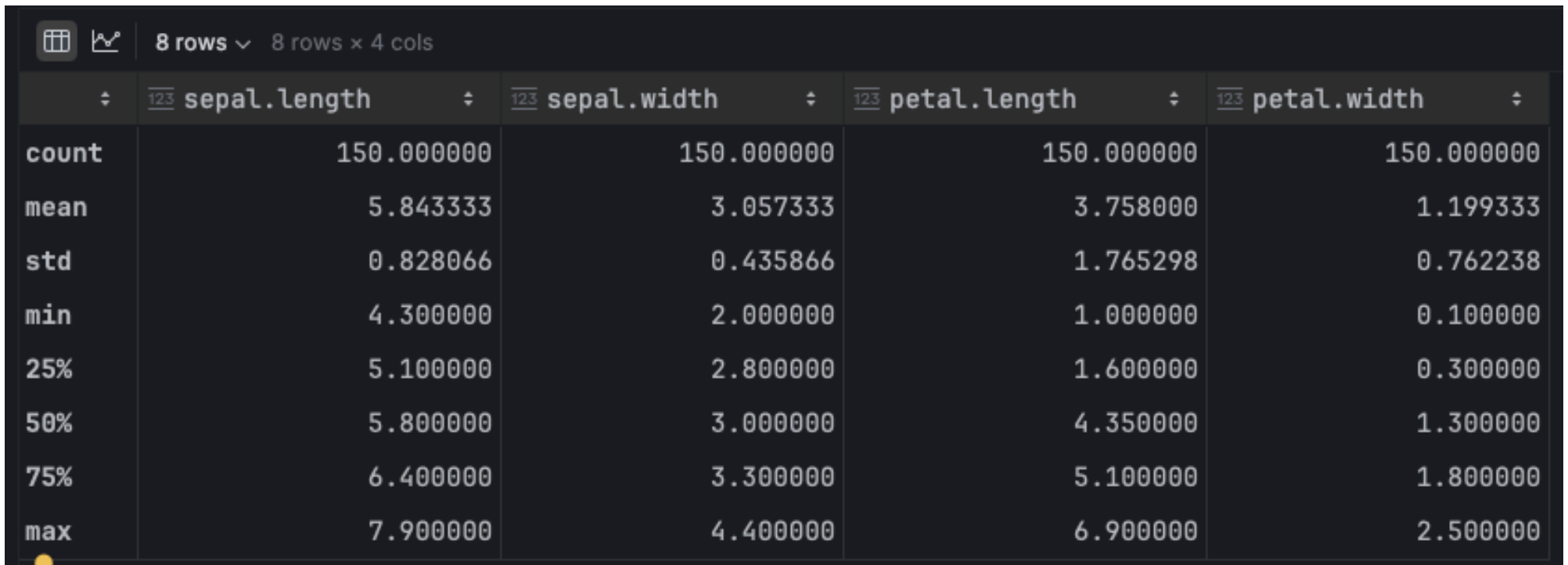
Descriptive statistics summarize the main characteristics of a numerical variable.

Common metrics include:

- **Mean**
- **Median**
- **Standard Deviation**
- **Minimum and Maximum**
- **Quartiles**

These statistics help us understand the **central tendency and spread** of the data.

Using pandas.describe()



8 rows ▾ 8 rows × 4 cols

÷	<u>123</u> sepal.length	÷	<u>123</u> sepal.width	÷	<u>123</u> petal.length	÷	<u>123</u> petal.width	÷
count	150.000000		150.000000		150.000000		150.000000	
mean	5.843333		3.057333		3.758000		1.199333	
std	0.828066		0.435866		1.765298		0.762238	
min	4.300000		2.000000		1.000000		0.100000	
25%	5.100000		2.800000		1.600000		0.300000	
50%	5.800000		3.000000		4.350000		1.300000	
75%	6.400000		3.300000		5.100000		1.800000	
max	7.900000		4.400000		6.900000		2.500000	

Mean vs Median

Both measure the **central value** of a distribution.

Mean

- Average of all values
- Sensitive to extreme values

Median

- Middle value when data is sorted
- More robust to outliers

Skewness

Skewness measures the **asymmetry of a distribution**.

Types of skewness:

Symmetric distribution

- Mean $\approx$ Median

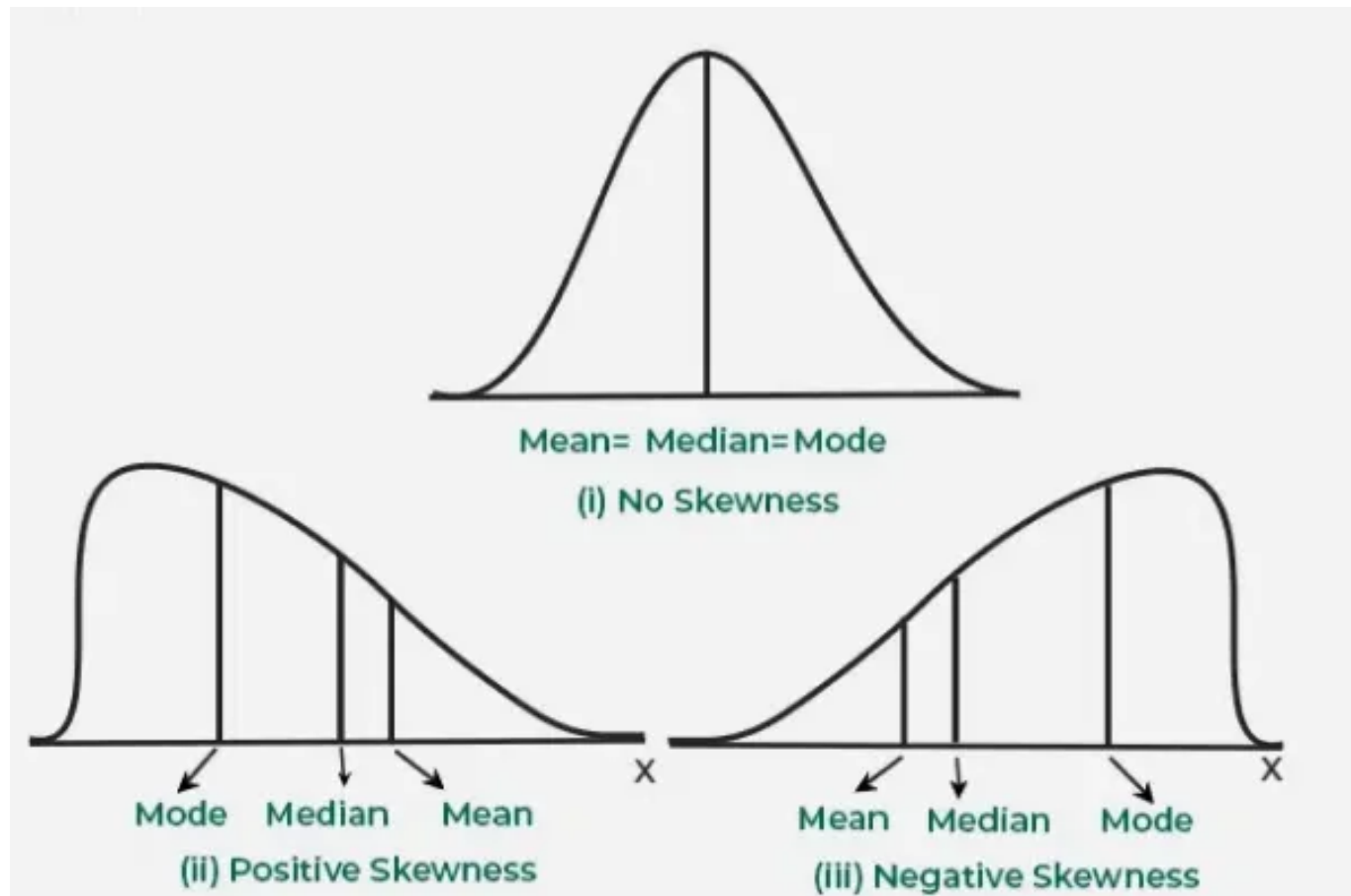
Positive skew (right skew)

- Long tail on the right

Negative skew (left skew)

- Long tail on the left

Skewness



<https://www.geeksforgeeks.org/data-science/skewness-measures-and-interpretation/>

Kurtosis

Kurtosis describes the weight of the tails of a distribution, indicating how frequently extreme values occur.

Common interpretations

Mesokurtic (Kurtosis ≈ 3 | Excess Kurtosis ≈ 0)

- Normal distribution
- Moderate tails

Leptokurtic (Kurtosis > 3 | Excess Kurtosis > 0)

- Heavy tails
- More extreme values

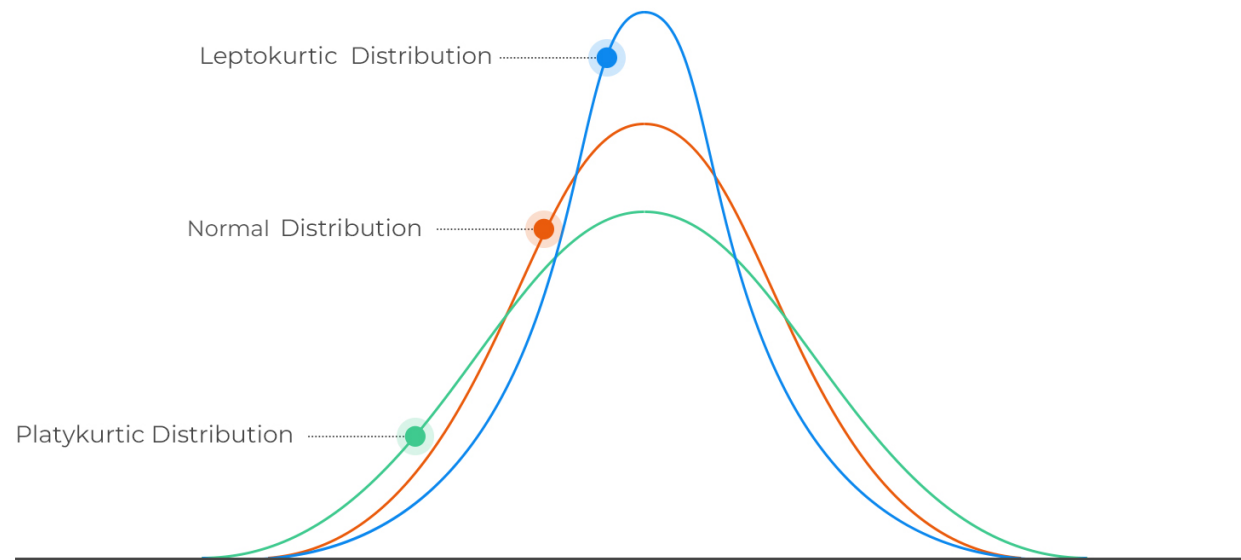
Platykurtic (Kurtosis < 3 | Excess Kurtosis < 0)

- Lighter tails
- Fewer extreme values

Kurtosis



Kurtosis



Categorical Variable Analysis

For categorical variables, we are mainly interested in:

- **Frequency of each category**
- **Relative proportion of categories**
- **Distribution across categories**

This helps us understand how the data is **structured across groups**.

Category Frequencies

A common way to explore categorical variables is by counting occurrences.

In pandas:

```
df["variable"].value_counts()
```

This shows:

- The number of observations per category
- Which categories are most common

Cardinality

Cardinality refers to the **number of unique categories** in a variable.

Examples:

Low cardinality

- Gender
- Payment type

High cardinality

- Customer ID
- Product ID
- City names

Grouping Rare Categories

Some categories may appear **very rarely** in the dataset.

A common strategy is to group them into a single category:

"Other"

Example:

MS Zoning	
RL	2273
RM	462
FV	139
RH	27
C (all)	25
I (all)	2
A (agr)	2

MS Zoning	
RL	77.576792
RM	15.767918
FV	4.744027
RH	0.921502
C (all)	0.853242
I (all)	0.068259
A (agr)	0.068259

Why Data Visualization?

Visualizations help us:

- Understand the **distribution of variables**
- Detect **patterns and trends**
- Identify **anomalies and outliers**
- Communicate insights clearly

Graphs often reveal patterns that are **not obvious in tables or statistics.**

Histogram

A **histogram** shows the **distribution of a numerical variable**.

It works by:

- Dividing the data into **bins**
- Counting how many observations fall into each bin

Histograms help us identify:

- Distribution shape
- Skewness
- Potential outliers

Boxplot

A **boxplot** summarizes the distribution of a variable using quartiles.

Key components:

- **Q1 (25th percentile)**
- **Median (50th percentile)**
- **Q3 (75th percentile)**
- **Interquartile Range (IQR)**
- **Outliers**

Boxplots are very useful for detecting **extreme values**.

Bar Plot

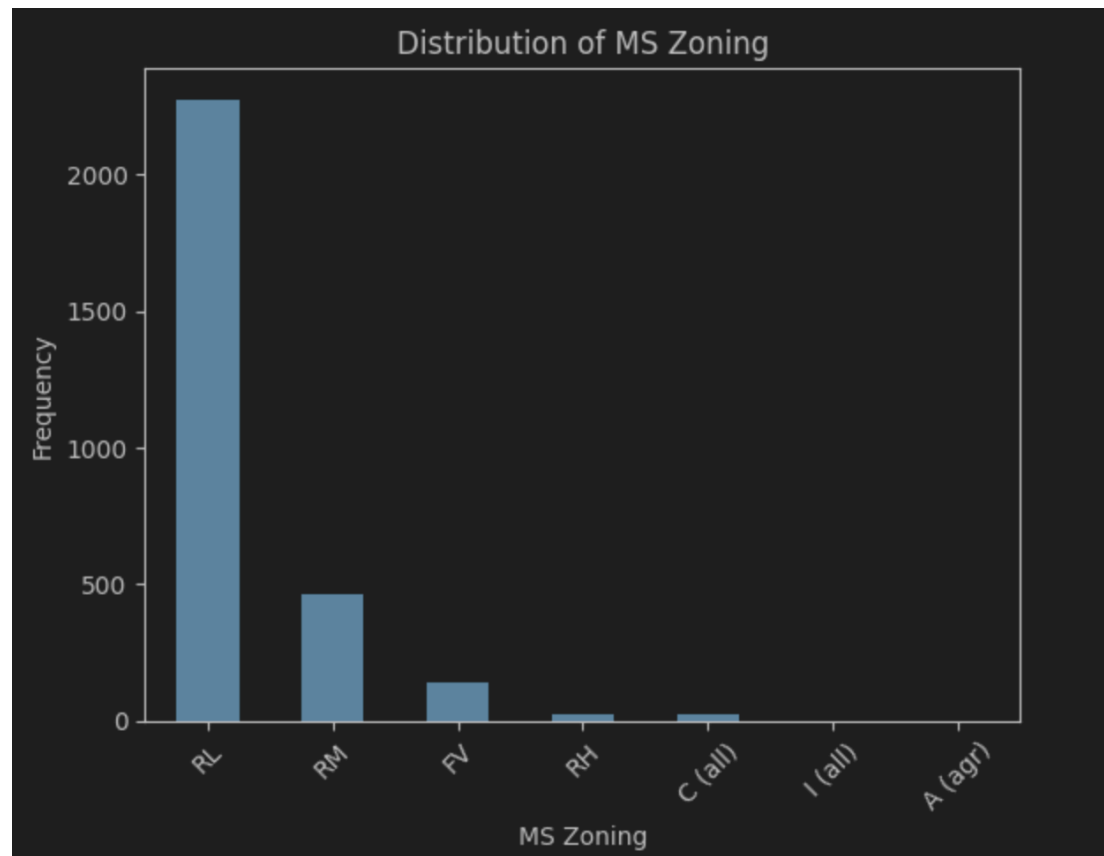
A **bar plot** is commonly used for **categorical variables**.

It shows:

- Categories on the x-axis
- Frequency or proportion on the y-axis

Bar plots help us quickly see which categories are **most common**.

Bar Plot



Obrigado!